



A Week in the Life of an Architect

Frank Buschmann

IN THE FIRST installment of the Pragmatic Architect column, I argued that architects shouldn't be disconnected from their projects and deliberate development practices. I also argued that architects should neither be blinded by the "beauty" of their own designs nor believe that their role makes them better programmers than the rest of the team.¹ Instead, architects should strive to

- express designs and their technical implications in terms of business stakeholder interests,
- guide development by providing architectural vision and driving design decisions with system-wide im-

However, we haven't yet explored one of my first column's observations: "It's a real challenge for architects to clearly understand at what level of abstraction they must be at a given time, in a given development phase, or during a given activity, for a given stakeholder or task. [...] It's a challenge for them to keep all levels in sync and balanced."¹

Anywhere, Anytime

From a bird's-eye perspective, the solution to this challenge is surprisingly simple: All activities—from business and architectural strategy to programming—should happen more or less simultaneously. They should occur in bite-sized chunks so that no activity is

while working on a product line for industrial automation systems. My role was to mentor the architect team. Development largely followed agile and lean practices, and the actual project phase was architectural elaboration. We were developing a walking skeleton that focused on essential functional scenarios and their associated nonfunctional quality.

Monday Morning

Analyzed the results of tests running over the weekend. Functional tests are all green, but some quality aspects don't meet expectations. Met with the responsible architect to localize the bottlenecks and explore potential measures for their resolution.

Monday Afternoon

Joined the test team to design and realize additional tests for guiding the resolution of the quality bottlenecks identified this morning. Also participated in a code walkthrough of the quality tests for newly developed functionality.

Monday Evening

Prepared the status report for tomorrow's project management meeting.

Tuesday Morning

At project management meeting, discussed the project's status. Proposed that next activities should focus on quality, not new functionality. Proposal

Architects should neither be blinded by the "beauty" of their own designs nor believe that they're the best.

- understand that code matters!

Subsequent columns have explored design and development methods, techniques, and practices that help architects design sustainable systems that meet their business objectives.

predominant, no stakeholder group feels neglected, and design and code are always under control concerning both progress and quality. This ambition, however, is easier to postulate than realize. So let's investigate.

Here's the compressed "diary" of a real week in my life as an architect

was accepted, yet we had a lengthy discussion with the “feature advocates.”

Tuesday Afternoon

Participated in a code walkthrough for the reference implementation of a quality-relevant architectural concept. We did some refactoring but concluded that the code was of sufficient quality for integration with the walking skeleton. Scheduled a workshop with the test team to develop a set of tests to ensure performance and robustness of the concept.

Wednesday Morning

Took half a day off to ride my (dressage) horse Eddi. He doesn't care about computing. Felt really good.

Wednesday Afternoon

Strategy meeting with management to discuss how to set up a product-line engineering business and a development process for the current product line. We focused on the core-asset business model, product-line scoping, and requirements management.

Thursday Morning

Participated in the team's stand-up meeting. We focused on the nonfunctional quality of the walking skeleton. Analyzed the results of the mid-week quality tests. Quality improved, but we haven't resolved all the deficiencies identified on Monday. A new bottleneck was identified. Aligned with project manager on the roadmap for quality improvement.

Thursday Afternoon

Participated in the realization of a reference implementation for an architectural concept not yet integrated with the walking skeleton: pair programming, refactoring, and unit testing.

Thursday Evening

Prepared material for an architectural strategy workshop with top management.

Friday Morning

Joined the weekly architecture meeting. Discussed consequences of recent quality tests on the architecture as well as open design issues relevant to the walking skeleton.

Friday Afternoon

Workshop with top management on architectural strategy for automation systems. We focused on innovation goals and a platform strategy.

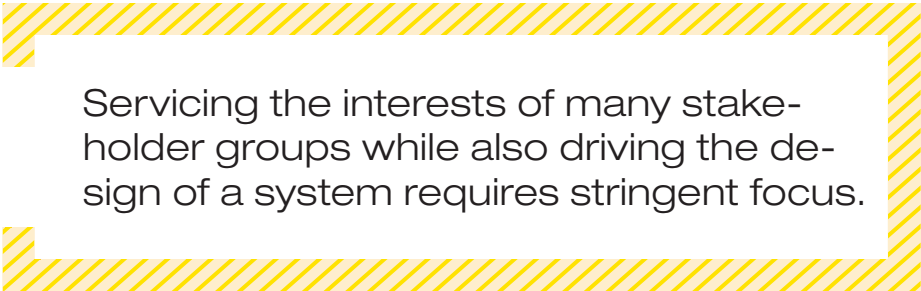
Focus Is Essential

Servicing the interests of many stakeholder groups while also driving the

process of a system's design, they can't seriously address more than one or two of them at one time—and then only bite by bite. One focus per week seems like an appropriate bite size to me. It's big enough to achieve visible progress, yet small enough to address all relevant aspects of a given project phase within a reasonable timeframe. Themes beyond this focus aren't ignored but observed and added to the backlog rather than immediately addressed—until their time comes.

Where Do You Want to Be?

Architects should choose a goal for



Servicing the interests of many stakeholder groups while also driving the design of a system requires stringent focus.

design of a system requires stringent focus. During the week that my diary sketches, I was focused on the nonfunctional quality of the scenarios the walking skeleton implemented and the presence of relevant architectural design concepts that referenced implementation quality. Focus can differ depending on a project's phase, status, and progress, but each week, increment, or iteration must have a dedicated theme.

Why? If architects, including myself, try to address too many topics in parallel, they're likely to get overloaded while actually achieving very little. They're everywhere but not where they need to be.² Their impact is limited. Developers often don't accept architects who act this way, and the more that architects' busyness mismatches the project's progress, the more that business stakeholders tend to doubt their competence.

Although architects must address all aspects of relevance to ensure the suc-

cess of each week or project phase. Working on a specific topic is only worthwhile if it achieves something that contributes to the project's progress. Otherwise, architects tend to waste time on things that are technically interesting but have limited relevance to the situation at hand. I had two goals for the week I outlined: improving the performance of the most critical scenarios in the walking skeleton and getting the go ahead for the integration of an important architectural concept. I didn't fully achieve both goals—some performance issues weren't completely resolved. Yet, I can only say this because I had a concrete goal. These goals can correspond to specific requirements or milestones of high-priority architectural issues in the backlog. Goals drive all my activities. I'm less concerned about the way to achieve the goals than achieving the goals themselves. I'd rather have the

journey to achievement be pragmatic and straight than theoretically and methodologically correct.

Architecting by Walking Around

If you observe how pragmatic architects design, you'll notice that they spend most of their time with the people who must live with their decisions: project and business managers, developers, testers, and so on. They spend less time in meetings and more time in workshops that directly elaborate the architecture. You can see this practice in the diary where I was "just" the architectural mentor, not the responsible architect. Still, architects must allocate time for the initial design creation. But the result is more of a vision or a rough upfront design that guides subsequent work with the development team, not a fully developed blueprint that requires strict enforcement.

The architect then refines the initial vision in direct cooperation with the development team and stakeholders. His presence indicates that he takes their interests, concerns, and feedback seriously; his cooperation encourages

and empowers them to actively shape the system's design. Stakeholders are no longer receivers of the architecture but contributors to the final result. Decision chains and feedback loops concerning the architecture move closer together in both space and time. The architectural vision is gradually stiffened toward a walking skeleton and beyond.^{3,4} The process isn't necessarily straight; it often requires endurance over multiple iterations. But it's worth it—if you succeed, you'll end up with a sustainable architecture that receives buy-in from all parties.

The lesson we can extract from this reflection is direct: the only feasible way for architects to balance all their duties is through leadership. They must have a clear architectural vision for the system under development. During its realization, they must keep the focus on the success-critical aspects—not on all aspects at once but on one or two at a time. Each aspect should remain under focus long enough to achieve impact but short enough to

not lose control over other aspects that require attention. Architects must continuously work toward measurable and realistic goals—short and long term—in direct cooperation with all involved stakeholder groups.

Leadership, paired with proficiency in software engineering methods, practices, and technologies, is the key to an architect's success. You can't always "win," but you can significantly minimize failure. ☞

References

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